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TASK	DNS IN DETAIL REPORT

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DNS IN DETAIL REPORT

Introduction

This report documents what I learned from the TryHackMe DNS in Detail module as part of my Cyber Shujaa training. The module focused on understanding how the Domain Name System works, how domain names are structured, and how different DNS records are used to support internet communication.

1.1 What is DNS?

In this section, I learned that DNS (Domain Name System) translates human-readable domain names into IP addresses that computers use to communicate on the internet. DNS makes it easier to access websites without having to remember complex numerical IP addresses.

1.2 Domain Hierarchy

In this section, I learned about the domain name hierarchy, including Top-Level Domains (TLDs), Second-Level Domains, and subdomains. I learned the difference between generic and country-code TLDs, how second-level domains are structured, and how subdomains are used to organize services within a domain.

Question1: What is the maximum length of a subdomain?

Answer: **63**

Question2: Which of the following characters cannot be used in a subdomain (3 b _ -)?

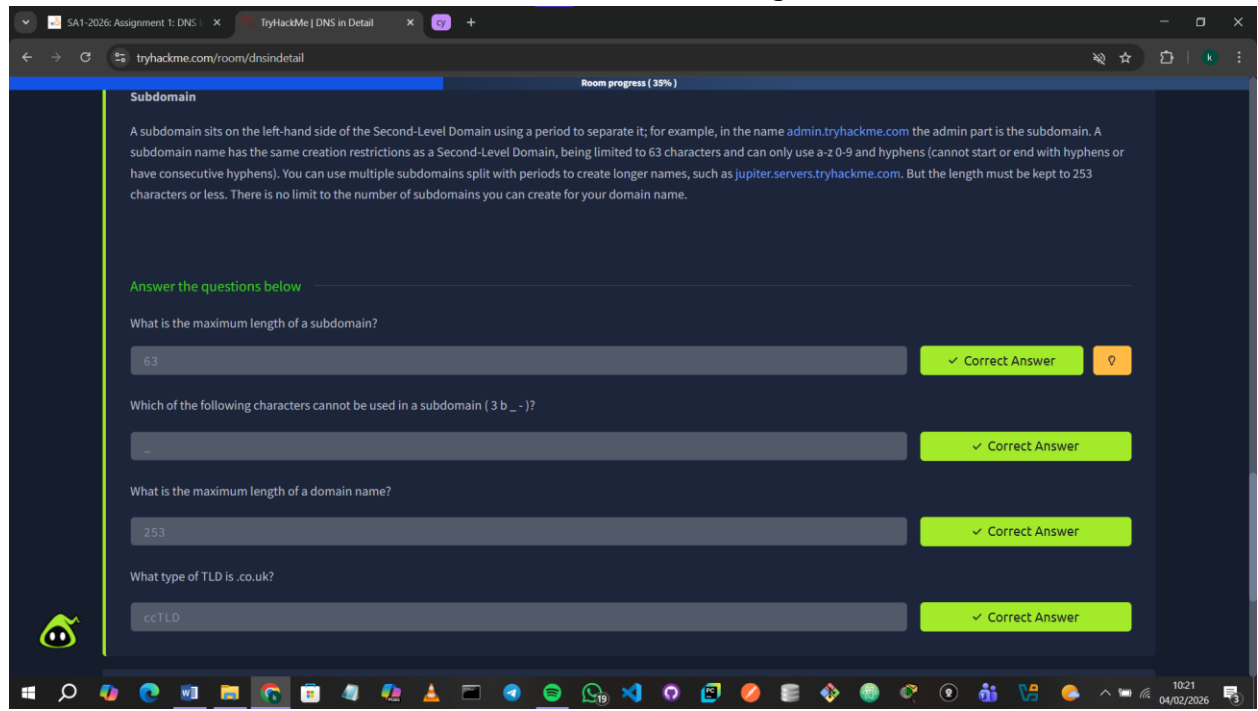
Answer: **_**. A subdomain can only use a-z 0-9 and hyphens.

Question3: What is the maximum length of a domain name?

Answer: **253**

Question4: What type of TLD is .co.uk?

Answer: **ccTLD**. .uk is used for sites based in the United Kingdom



1.3 Record Types

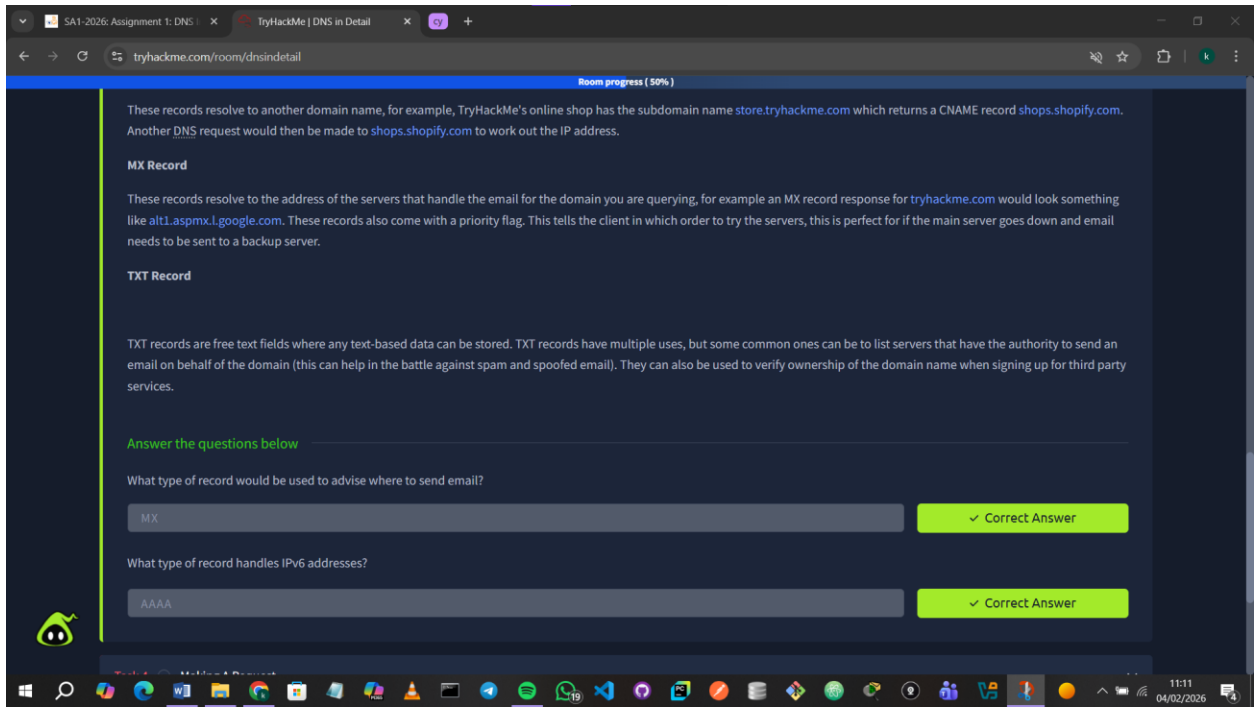
In this section, I learned about common DNS record types and their purposes. These include A and AAAA records for IPv4 and IPv6 addresses, CNAME records for domain aliases, MX records for email routing, and TXT records for storing verification and email security information.

Question1: What type of record would be used to advise where to send an email?

Answer: **MX**

Question2: What type of record handles IPv6 addresses?

Answer: **AAAA**



1.4 Making a Request

In this section, I learned how a DNS request is processed from start to finish. The request checks local and recursive DNS caches first, then queries root servers, TLD servers, and finally the authoritative name server. I also learned how DNS caching and TTL values improve performance by reducing repeated lookups.

Question1: What field specifies how long a DNS record should be cached for?

Answer: **TTL**

Question2: What type of DNS Server is usually provided by your ISP?

Answer: **Recursive**

Question3: What type of server holds all the records for a domain?

Answer: **Authoritative**

The screenshot shows a web browser window with the URL `tryhackme.com/room/dnsindetail`. The page has a dark theme and a blue header bar that says "Room progress (73%)".

Text content on the page:

- 4. The TLD server holds records for where to find the authoritative server to answer the DNS request. The authoritative server is often also known as the nameserver for the domain. For example, the name server for `tryhackme.com` is `kip.ns.cloudflare.com` and `uma.ns.cloudflare.com`. You'll often find multiple nameservers for a domain name to act as a backup in case one goes down.
- 5. An authoritative DNS server is the server that is responsible for storing the DNS records for a particular domain name and where any updates to your domain name DNS records would be made. Depending on the record type, the DNS record is then sent back to the Recursive DNS Server, where a local copy will be cached for future requests and then relayed back to the original client that made the request. DNS records all come with a TTL (Time To Live) value. This value is a number represented in seconds that the response should be saved for locally until you have to look it up again. Caching saves on having to make a DNS request every time you communicate with a server.

Diagram: A blue globe icon is connected by a green arrow to a server rack icon labeled "DNS". Another green arrow points from the server rack to the text "4 Authoritative".

Section: Answer the questions below

What field specifies how long a DNS record should be cached for?

✓ Correct Answer

What type of DNS Server is usually provided by your ISP?

✓ Correct Answer

What type of server holds all the records for a domain?

✓ Correct Answer

The Windows taskbar is visible at the bottom with various application icons and a system clock showing 11:25 on 04/02/2026.

1.5 Practical

In this section, I applied what I learned by performing practical DNS queries using an interactive tool.

Question1: What is the name of shop.website.thm?

Answer: **shops.myshopify.com**

Question2: What is the value of the TXT record of website.thm?

Answer: **THM{7012BBA60997F35A9516C2E16D2944FF}**

The screenshot shows a web browser window with the TryHackMe DNS room interface. The room progress is 85%. The interface includes a list of questions to be answered, a 'Check' button for each, and a 'Submit now' button at the bottom. The questions are:

- What is the CNAME of shop.website.thm? (Answer: shops.myshopify.com)
- What is the value of the TXT record of website.thm? (Answer: THM{7012BBA60997F35A9516C2E16D2944FF})
- What is the numerical priority value for the MX record? (Answer: 30)
- What is the IP address for the A record of www.website.thm? (Answer: 10.10.10.10)

Below the questions is a section for recommending the room to others, with a rating from 1 to 10. The terminal output on the right shows the following commands and results:

```
user@thm:~$ nslookup --type=CNAME shop.website.thm
Server: 127.0.0.53
Address: 127.0.0.53#53

Non-authoritative answer:
shop.website.thm canonical name = shops.myshopify.com

user@thm:~$ nslookup --type=TXT website.thm
Server: 127.0.0.53
Address: 127.0.0.53#53

Non-authoritative answer:
website.thm text = "THM{7012BBA60997F35A9516C2E16D2944FF}"

user@thm:~$ nslookup website.thm
```

Question3: What is the numerical priority value for the MX record?

Answer: **30**

Question4: What is the IP address for the A record of www.website.thm?

Answer: **10.10.10.10**

The screenshot shows the TryHackMe DNS room interface with the room progress at 92%. The questions are now marked as 'Correct Answer'. The terminal output on the right shows the following commands and results:

```
user@thm:~$ nslookup --type=MX website.thm
Server: 127.0.0.53
Address: 127.0.0.53#53

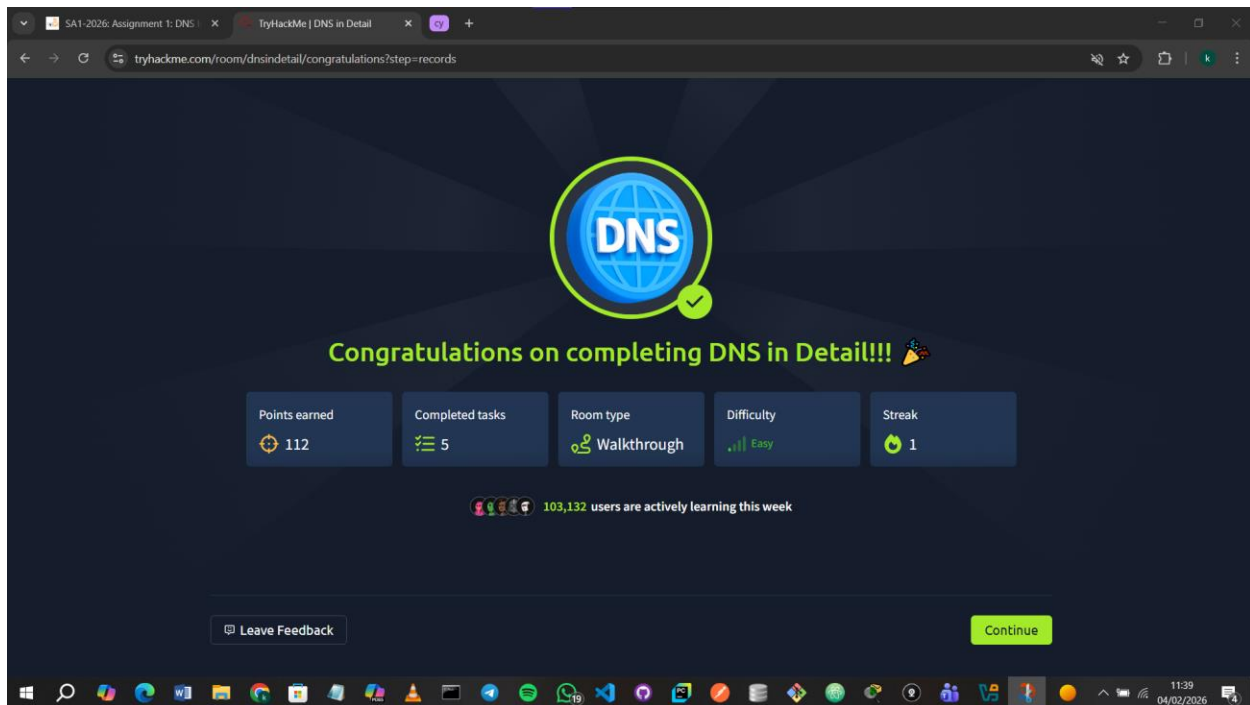
Non-authoritative answer:
website.thm mail exchanger = 30 alt4.aspmx.l.google.com

user@thm:~$ nslookup --type=A www.website.thm
Server: 127.0.0.53
Address: 127.0.0.53#53

Non-authoritative answer:
Name: www.website.thm
Address: 10.10.10.10

user@thm:~$ nslookup website.thm
```

A notification bubble in the top right corner of the terminal window states: "2/4 Rooms completed! Levelling up in progress..."



Conclusion

Through this module, I gained a solid understanding of DNS concepts, including domain hierarchy, DNS record types, and how DNS requests are processed. By completing the practical tasks, I was able to apply this knowledge to perform real DNS queries, strengthening my ability to analyze and understand DNS related information in a cybersecurity context.